

Tyson L. Swetnam, Ph.D.

Research Computing, Data Platforms & Applied AI Leader

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SUMMARY

Technical leader with 15+ years building and operating cloud, HPC, and data infrastructure for scientific research. Director of the University of New Mexico's Center for Advanced Research Computing (CARC). Former Co-PI of CyVerse, the NSF-funded national cyberinfrastructure for the life sciences. Track record of winning and delivering large federal programs, standing up platforms that serve thousands of researchers, and translating emerging technology (containers, cloud-native data, LLMs, and agentic AI) into production services and hands-on training.

CORE COMPETENCIES

Leadership	Research-computing strategy & operations; program management of multi-institution federal awards; budgets & proposals; team building; stakeholder & funder relations (NSF, USDA, NIH, NASA)
AI & ML	Large language models, agentic AI & automation, metadata-aware AI agents, knowledge graphs & ontologies, GPU computing, prompt engineering
Data platforms	Federated data mesh, lakehouse table formats (Apache Iceberg, DuckLake), iRODS, FAIR data & metadata curation, cloud-optimized & analysis-ready data
Infrastructure	HPC, public research clouds (Jetstream2, ACCESS), Docker, Singularity/Apptainer, Kubernetes, Linux, GitHub CI/CD
Languages & tools	Python, R, Bash, SQL, Jupyter, Google Earth Engine, GIS
Geospatial	Satellite, airborne & drone (sUAS) remote sensing; lidar & photogrammetry; environmental monitoring

EXPERIENCE

Director, Center for Advanced Research Computing 7/2026 – Present
University of New Mexico • also Associate Professor of Computer Science Albuquerque, NM

- Lead UNM's central HPC, storage, and research-computing center, which serves faculty and students across the state's flagship university.
- Setting direction for open-access research data storage and for making advanced computing and AI accessible to all New Mexico researchers.
- PI of **NSF MESA** (\$4.6M, 2026–2029), a NAIRR research-infrastructure platform in which metadata-aware AI agents describe, organize, and connect scientific data across fields.

Research Associate Professor 7/2023 – 6/2026
University of Arizona, Institute for Artificial Intelligence & Society Tucson, AZ

- Led the University of Arizona cyberinfrastructure work for two NSF synthesis centers: ESIIL (\$20M program; \$1.46M UA scope) and NCEMS (\$1.93M UA scope).
- Senior personnel on the USDA–NSF AI Institute for Resilient Agriculture (AIIRA, \$20M) and on the Data Management Core of the NIH Superfund DUST Center (\$14.8M).
- Created and taught LLM and agentic-AI workshops and course materials (GPT-101; AI Automation & Agents) for researchers, staff, and students.

Research Assistant Professor & Data Scientist, CyVerse 9/2016 – 6/2023
University of Arizona, BIO5 Institute Tucson, AZ

- Co-PI of **CyVerse** (NSF, \$15.2M), national cyberinfrastructure providing data storage, cloud analysis, and containerized apps to the life-science research community.

- Built and ran Container Camp and Foundational Open Science Skills training, teaching Docker, Kubernetes, Singularity, and reproducible workflows to researchers nationwide.
- Delivered scalable GPU and ML pipelines for crop phenomics (PhytoOracle) and drone data, and secured TACC Frontera and Longhorn GPU allocations.
- Senior personnel on the NSF HydroGEN (\$5M) and Groundwater Data Platform (\$1M) machine-learning water-resource projects.

Associate Research Scientist / Postdoctoral Researcher, *University of Arizona & University of Utah* 2014 – 2016

Fire Management Specialist, *USDA Forest Service, Coronado National Forest* 2008 – 2012

Forestry Technician (Wildland Firefighter), *U.S. Dept. of the Interior, Saguaro National Park* 2002 – 2005

EDUCATION

Ph.D., Natural Resources (minor: Remote Sensing & Spatial Analysis), *University of Arizona* 2013

M.S., Natural Resources (GIS Technical Certificate), *University of Arizona* 2006

B.S., Ecology & Evolutionary Biology, *University of Arizona* 2002

SELECTED IMPACT & RECOGNITION

- PI, Co-PI, or senior personnel on 25+ funded projects, including four programs of \$14M–\$20M each.
- 40 peer-reviewed publications; 2,000+ citations (h-index 26, Google Scholar).
- Best Short Paper, Workforce Development, PEARC '23; ABOR TRIF Eighteenth Mile Award (\$141K, 2023) for AEOLUS, a drone-data laboratory.
- Grant review panelist for NSF CISE, NSF DBI, and USDA NIFA; member of the NSF CI Compass Cloud Infrastructure Working Group; certified Carpentries instructor.
- Invited speaker at TACC, NASA JPL, NVIDIA GPU Cloud, the Santa Fe Institute, the University of Cambridge, and NSF–CNPq (Brazil).

SELECTED PUBLICATIONS

Swetnam, T.L., P.B. Antin, R. Bartelme, et al. (2024) CyVerse: Cyberinfrastructure for open science. *PLOS Computational Biology*. doi:10.1371/journal.pcbi.1011270

Roy, A., E. Ward, I. Choi, et al. (2025) MDRepo – an open data warehouse for community-contributed molecular dynamics simulations of proteins. *Nucleic Acids Research* 53(D1). doi:10.1093/nar/gkae1109

Thessen, A., L. Cooper, **Swetnam, T.L.**, et al. (2023) Using knowledge graphs to infer gene expression in plants. *Frontiers in Artificial Intelligence* 6:92. doi:10.3389/frai.2023.1201002

Skidmore, E., M. Cosi, **Swetnam, T.L.**, et al. (2023) Cloud computing for research and education gets a sweet upgrade with CACAO. *PEARC '23*. ACM. doi:10.1145/3569951.3597555

Gonzalez, E.M., A. Zarei, N. Hendler, et al. (2023) PhytoOracle: Scalable, modular phenomics data processing pipelines. *Frontiers in Plant Science* 14. doi:10.1002/essoar.10508789.1